Fourier Transforms

DFT Properties

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Linearity

- $x_1[n] \xleftrightarrow{L_1-point\ DFT} X_1[k]$
- $x_2[n] \xleftrightarrow{L_2-point\ DFT} X_2[k]$
- $A_1x_1[n] + A_2x_2[n] \xleftrightarrow{N=\max(L_1,L_2)-point\ DFT} A_1X_1[k] + A_2X_2[k]$

Circular Shift in Time

- $x[n] \xleftrightarrow{N-point\ DFT} X[k]$
- $x[(n-m)_N] \xleftrightarrow{N-point\ DFT} X[k]e^{-j\frac{2\pi k}{N}m}$

Modulation (Circular Shift in Frequency)

- $x[n] \xleftrightarrow{N-point\ DFT} X[k]$

- $x[n]e^{j\frac{2\pi l}{N}n} \xleftrightarrow{N-point\ DFT} X[(k-l)_N]$

Time Reversal

- $x[n] \xleftrightarrow{N-point\ DFT} X[k]$
- $x[(-n)_N] \xleftrightarrow{N-point\ DFT} X[(-k)_N]$

Complex Conjugation

- $x[n] \xleftrightarrow{N-point\ DFT} X[k]$

- $x^*[n] \xleftrightarrow{N-point\ DFT} X^*[(-k)_N]$

- **If *x[n]* is a real signal,**
  - $x[n] = x^*[n]$
  - $X[k] = X^*[(-k)_N]$

Question

Suppose x[n] is a real discrete-time signal of length 8. Which of the following is correct regarding its 8-point DFT X[k]?

a) $X[1] = X[7]$

b) $X[2] = X[5]$

c) $X[3] = X^*[5]$

d) $X[4] = X^*[3]$

Circular Convolution

- $x_1[n] \xleftrightarrow{N-point\ DFT} X_1[k]$
- $x_2[n] \xleftrightarrow{N-point\ DFT} X_2[k]$
- $x_1[n] \textcircled{N} x_2[n] = \sum_{k=0}^{N-1} x_1[k] x_2[(n-k)_N] \xleftrightarrow{N-point\ DFT} X_1[k] X_2[k]$

DTFT, DFT, Linear and Circular Convolution

Assume $N \geq L = L_x + L_h - 1$

$$y[n] = x[n] * h[n] \quad \xleftrightarrow{\text{DTFT}} \quad Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

Replicate *y[n]* by *N*, take main period

Sample $Y(e^{j\omega})$ with period $2\pi/N$, take main period

$$\dot{y}[n] = \sum_{l=-\infty}^{\infty} y[n - lN], \quad = x[n] \,\textcircled{N}\, h[n], \quad n = 0,1,\ldots,N-1$$

$$\xleftrightarrow{N\text{-point DFT}}$$

$$Y[k] = X[k]H[k], \quad k = 0,1,\ldots,N-1$$

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